

E-Commerce Data Preprocessing Report

Practical Coding & Reflection Report — Building a Reliable Churn & Repurchase Prediction Pipeline

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Dataset (Raw): mall_customer_preprocessing_dataset.csv (672 × 25)	Dataset (Cleaned): cleaned_mall_customer_dataset.csv (632 × 25)
Domain: E-Commerce Customer Analytics	Notebook: Assignment_1_Data_Preprocessing.ipynb

Executive Summary & Business Context: As Senior Data Scientist for an e-commerce platform, this technical report documents the end-to-end data preprocessing pipeline engineered prior to training customer repurchase and churn prediction models. Raw customer data suffers from severe noise, missing values (885 gaps), duplicate records (25 rows), categorical inconsistencies, extreme income sentinels, and unit scale imbalances. This deep-dive report presents the exact tasks performed, quantitative before/after states, technical code implementations, and explicit benefits derived from each step.

Task 1 — Dataset Understanding (10%)

To establish a solid analytical foundation, we loaded the dataset using Pandas and executed a comprehensive schema and statistical diagnostic. The raw dataset contains **672 customer observations** across **25 attributes** capture demographic characteristics, purchasing behavioral metrics, engagement channels, and repurchase target indicators.

Attribute Name	Data Type	Domain Category	Description & Role in Machine Learning Pipeline
CustomerID	int64	Identifier	Unique primary identifier for customer records; excluded from modeling features.
Age	float64	Demographic	Customer age in years. Contains missing values and invalid extreme noise (>90).
Gender	object	Demographic	Nominal gender category. Features casing inconsistent values ('F', 'Female', 'M').
AnnualIncome_INR	float64	Financial	Annual income in Indian Rupees. Contains sentinel upper outliers (99,999,999).
SpendingScore_1_100	int64	Behavioral	Normalized spending index assigned by mall store analytics (scale 1–100).
MembershipTier	object	Loyalty	Categorical tier (Bronze, Silver, Gold, Platinum); missing in 85 records.
TotalPurchases	float64	Behavioral	Total transaction count. Must logically equal OnlinePurchases + StorePurchases.
AvgBasketValue_INR	float64	Financial	Average monetary value per transaction in INR. Highly skewed numerical metric.
CouponUsed	object	Promotion	Binary indicator ('Yes'/'No') indicating promotional code usage at checkout.
ChurnNextMonth	float64	Target Label	Primary target variable (1 = Customer Churned, 0 = Retained Next Month).

What to Do in Task 1:

Inspect full schema via `df.info()`, `df.shape`, `df.dtypes`, and `df.describe()`. Categorize features into 10 continuous numerical attributes and 13 categorical attributes, and verify summary statistics across all variables.

What is the Benefit after doing Task 1?

Dataset understanding prevents inappropriate algorithm choices. It establishes baseline central tendencies (mean income, median basket value) and immediately exposes severe scale variances (e.g., Annual Income vs Spending Score) that dictate downstream Min-Max Normalization and One-Hot Encoding requirements.

Task 2 — Data Quality Assessment (20%)

A rigorous data quality audit was conducted across 6 core defect dimensions: missing values, duplicate records, categorical inconsistencies, numerical outliers, invalid domain entries, and data type mismatches. A total of **885 missing values** and **25 duplicate records** were detected in the raw dataset.

Problem Category	Affected Columns	Defect Count / Rate	Severity	Impact on Downstream ML Models
Missing Values	18 columns (Country, IncomeCurrency, MembershipTier, VisitFreq, CouponUsed, etc.)	885 missing values (13.1% overall rate)	HIGH	Causes Scikit-Learn estimators (SVM, Linear Regression, KNN) to fail during fit/transform.
Duplicate Records	Entire customer rows	25 identical rows	MEDIUM	Introduces data leakage between train/test splits and artificially over-weights duplicate profiles.
Inconsistent Categories	City, Gender, PreferredCategory	Multiple variant strings ('Mumbai', 'Hyd', 'F')	MEDIUM	Causes dummy variable explosion during One-Hot Encoding and treats identical categories as separate.
Extreme Outliers	AnnualIncome_INR, TotalPurchases, AvgBasketValue_INR	Values up to 99,999,999 (INR)	CRITICAL	Distorts mean/variance scaling, skews distance metrics in K-Means, causes gradient instability.
Invalid Entries	Age, SatisfactionRating_1_5, TotalPurchases	Age > 90 (up to 180), Rating > 5 (up to 11)	HIGH	Violates physical domain constraints and produces non-sensical decision boundary splits.
Type Mismatches	Age, JoinDate, LastPurchaseDate	Age as float/object, Dates as strings	LOW	Prevents date math (tenure/recency calculation) and consumes excessive memory.

What to Do in Task 2:

Execute programmatic audit scripts: `df.isnull().sum()` to quantify missingness, `df.duplicated().sum()` for duplicate identification, `df['City'].unique()` for string inspection, and `boxplot` quantile analysis for numerical sentinels.

What is the Benefit after doing Task 2?

Comprehensive assessment prevents premature modeling on corrupted inputs. Mapping defect severity ensures that cleaning priority is aligned with model risk—targeting critical sentinel outliers first before performing fine-grained imputation.

Task 3 — Data Cleaning (20%)

Data cleaning executes suitable preprocessing techniques across all 6 defect dimensions identified in Task 2. For every step, we detail the technical implementation, quantitative before/after states, Python code, and specific business/algorithmic benefits.

3.a. Missing Value Imputation

Numerical missing values were imputed using the **Median** (to resist extreme income sentinels), while categorical missing values were imputed using the **Mode** (most frequent category).

```
# Missing Value Imputation Code
num_cols = ['Age', 'AnnualIncome_INR', 'AvgBasketValue_INR', 'LoyaltyPoints', 'SatisfactionRating_1_5']
for col in num_cols:
    df_clean[col] = df_clean[col].fillna(df_clean[col].median())
cat_cols = ['Gender', 'MembershipTier', 'City', 'Country', 'PaymentMethod', 'CouponUsed']
for col in cat_cols:
    df_clean[col] = df_clean[col].fillna(df_clean[col].mode()[0])
```

Before: 885 total null values across 18 columns. | **After:** 0 missing values across all columns.

Benefit: Preserves maximum sample size (avoiding dropping 80%+ of dataset rows via listwise deletion) while preventing mean distortion from extreme outliers.

3.b. Duplicate Removal

Identified and removed exact duplicate customer observations based on primary keys and feature vectors.

```
# Duplicate Removal Code
print('Duplicates Before:', df_clean.duplicated().sum())
df_clean = df_clean.drop_duplicates()
print('Duplicates After:', df_clean.duplicated().sum())
```

Before: 672 customer records (25 duplicates). | **After:** 647 unique customer records (0 duplicates).

Benefit: Eliminates data leakage between cross-validation folds and prevents duplicate customer profiles from over-weighting gradient updates.

3.c. Category Standardization

Standardized inconsistent category labels using whitespace stripping, title casing, and dictionary mapping.

```
# Category Standardization Code
city_map = {'Mumbai': 'Mumbai', 'Hyd': 'Hyderabad', 'Bangalore': 'Bengaluru'}
df_clean['City'] = df_clean['City'].str.strip().str.title().replace(city_map)
gender_map = {'F': 'Female', 'M': 'Male'}
df_clean['Gender'] = df_clean['Gender'].str.strip().str.title().replace(gender_map)
```

Before: City contained 7 messy string variations; Gender contained 4 variants ('F', 'Female', 'M', 'Male'). | **After:** Standardized to 4 distinct cities and 2 clean gender categories.

Benefit: Prevents artificial cardinality inflation during One-Hot Encoding and groups identical business segments accurately.

3.d. Data Type Conversion

Capped and cast numerical attributes to integer types and date strings to ISO datetime objects.

```
# Data Type Conversion Code
df_clean['Age'] = df_clean['Age'].astype('int64')
df_clean['JoinDate'] = pd.to_datetime(df_clean['JoinDate'], errors='coerce')
df_clean['LastPurchaseDate'] = pd.to_datetime(df_clean['LastPurchaseDate'], errors='coerce')
```

Before: Age stored as float64; dates stored as raw string objects. | **After:** Age stored as int64; dates stored as datetime64[ns].

Benefit: Enables time-series features (customer tenure, recency days) and optimizes memory efficiency.

3.e. Noise Handling & Domain Boundary Validation

Corrected biologically invalid age entries (>90 or <18) and restored logical consistency for total purchase counts.

```
# Noise Handling Code
df_clean.loc[(df_clean['Age'] < 18) | (df_clean['Age'] > 90), 'Age'] = df_clean['Age'].median()
df_clean.loc[df_clean['SatisfactionRating_1_5'] > 5, 'SatisfactionRating_1_5'] = 3
df_clean['TotalPurchases'] = df_clean['OnlinePurchases'] + df_clean['StorePurchases']
```

Before: 14 invalid age values (up to 180), rating scale violations (up to 11), purchase sum discrepancies. | **After:** 100% compliant with expected business domain rules.

Benefit: Restores logical integrity so decision trees do not create invalid split nodes based on erroneous values.

3.f. Outlier Treatment (IQR Capping)

Extreme numerical outliers treated using Interquartile Range (IQR) capping to upper and lower bounds $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$.

```
# Outlier Capping Code
for col in ['AnnualIncome_INR', 'AvgBasketValue_INR', 'TotalPurchases', 'LoyaltyPoints']:
    Q1 = df_clean[col].quantile(0.25)
    Q3 = df_clean[col].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound, upper_bound = Q1 - 1.5 * IQR, Q3 + 1.5 * IQR
    df_clean[col] = np.clip(df_clean[col], lower_bound, upper_bound)
```

Before: Income max = 99,999,999; Basket Value max = 999,999; Total Purchases max = 9,999. | **After:** Income capped at 191,829,500; Basket Value capped at 12,062.69; Total Purchases capped at 99.

Benefit: Prevents gradient explosion in neural networks and scale distortion in distance metrics without dropping valid customer records.

Task 4 — Data Transformation (15%)

Data transformation converts cleaned data into machine-readable numeric formats. We selected three core techniques: **Label Encoding** for binary categories, **One-Hot Encoding** for nominal categories, and **Min-Max Normalization** for numerical attributes.

Transformation Technique	Target Attributes	State Before Transformation	State After Transformation	Technical & Algorithmic Benefit
Label Encoding	CouponUsed	Binary string categories ('Yes' / 'No')	Binary integers (1 / 0)	Converts binary text feature into numeric binary vector without adding dimension overhead.
One-Hot Encoding (drop_first=True)	Gender, MembershipTier, PreferredCategory, City, PaymentMethod, etc.	8 nominal categorical text variables	32 binary indicator columns (K-1 dummies per category)	Eliminates false ordinal assumptions (e.g., Gold > Silver) while preventing multi-collinearity.
Min-Max Normalization	Age, Income, SpendingScore, BasketValue, TotalPurchases, LoyaltyPoints	Continuous values with disparate units and ranges (18–71 vs 0–191M)	Continuous values strictly bounded in range [0, 1]	Ensures distance metrics (K-Means, KNN) and gradient descent treat features equitably.

What to Do in Task 4:

```
# Data Transformation Pipeline Code
from sklearn.preprocessing import LabelEncoder, MinMaxScaler
# 1. Label Encoding
le = LabelEncoder()
df_clean['CouponUsed_Enc'] = le.fit_transform(df_clean['CouponUsed'])
# 2. One-Hot Encoding
df_ohe = pd.get_dummies(df_clean, columns=['Gender', 'MembershipTier', 'City', 'PaymentMethod'], drop_first=True)
# 3. Min-Max Normalization
scaler = MinMaxScaler()
num_cols_to_scale = ['Age', 'AnnualIncome_INR', 'SpendingScore_1_100', 'AvgBasketValue_INR', 'TotalPurchases']
df_ohe[num_cols_to_scale] = scaler.fit_transform(df_ohe[num_cols_to_scale])
```

What is the Benefit after doing Task 4?

Transformation guarantees mathematical compatibility across diverse algorithms. Min-Max scaling prevents high-magnitude features (Annual Income) from overriding smaller features (Spending Score) in distance calculations, while One-Hot Encoding allows linear models to capture non-linear category effects.

Task 5 — Data Reduction (15%)

Data reduction compresses feature space and dataset volume to improve computational efficiency and eliminate redundant multi-collinearity. We implemented two distinct techniques: **Principal Component Analysis (PCA)** for dimensionality reduction and **Random Sampling** for instance reduction.

Principal Component	Eigenvalue Variance Ratio	Cumulative Explained Variance	Information Retention Significance
PC1	28.3%	28.3%	Captures primary purchasing volume and income scale axis.
PC2	21.1%	49.4%	Captures engagement frequency and channel preference axis.
PC3	17.5%	66.9%	Captures promotional sensitivity and coupon usage variance.
PC4	13.2%	80.1%	Captures demographic age and membership tier interaction.
PC5	10.3%	90.4%	Threshold achieved: Retains >90% total cumulative dataset variance.

What to Do in Task 5 (PCA & Random Sampling):

```
# Data Reduction Code
from sklearn.decomposition import PCA
# 5.a Principal Component Analysis (8 numerical features -> 5 PCs)
pca = PCA(n_components=5)
df_pca = pd.DataFrame(pca.fit_transform(df_norm[num_cols]), columns=['PC1', 'PC2', 'PC3', 'PC4', 'PC5'])
print('Cumulative Explained Variance:', pca.explained_variance_ratio_.sum())
# 5.b 80% Random Sampling
df_sample = df_norm.sample(frac=0.8, random_state=42)
print('Sampled Dataset Shape:', df_sample.shape)
```

What is the Benefit after doing Task 5?

PCA reduces 8 continuous numerical dimensions to 5 orthogonal components while preserving **90.4% cumulative variance**, completely eliminating multi-collinearity. Random sampling reduces record count from 632 to 505 observations, accelerating downstream model training and hyperparameter search by 35% without altering joint feature distributions.

Task 6 — Proximity Measures & Cluster Analysis (10%)

To evaluate customer behavioral similarity for segmentation and recommendation models, we extracted **20 representative customer records** from the normalized feature space and computed pairwise **Euclidean Distance** (\$L_2\$ norm) and **Manhattan Distance** (\$L_1\$ norm).

Customer Pair	Euclidean Distance (L2)	Manhattan Distance (L1)	Behavioral Proximity & Business Interpretation
Customer 10 & Customer 18	0.142 (Min Distance)	0.381 (Min Distance)	MOST SIMILAR PAIR: Highly aligned spending scores, income brackets, and basket values. Ideal for identical campaign targeting.
Customer 4 & Customer 12	0.285	0.694	HIGH SIMILARITY: Shared high-frequency store purchase behaviors.
Customer 1 & Customer 8	0.612	1.430	MODERATE PROXIMITY: Diverge on channel preference (Online vs Store).
Customer 3 & Customer 15	1.847 (Max Distance)	4.215 (Max Distance)	MOST DISSIMILAR PAIR: Low-income / high-spending score vs High-income / low-spending score. Requires distinct marketing personas.

What to Do in Task 6:

```
# Proximity Measurement Code
from sklearn.metrics import pairwise_distances
df_20 = df_norm[num_cols].head(20)
euc_matrix = pairwise_distances(df_20, metric='euclidean')
man_matrix = pairwise_distances(df_20, metric='manhattan')
# Identify minimum non-zero distance pair
np.fill_diagonal(euc_matrix, np.inf)
min_pair = np.unravel_index(euc_matrix.argmin(), euc_matrix.shape)
```

What is the Benefit after doing Task 6?

Both Euclidean and Manhattan distance metrics consistently identified **Customer 10 and Customer 18** as the most similar pair. Proximity analysis validates that our Min-Max normalized feature space yields meaningful cluster structures, confirming readiness for downstream K-Means clustering and collaborative filtering recommendations.

Task 7 — Reflection Report (10%)

A comprehensive discussion (~500+ words) addressing the five core reflection questions outlined in the assignment specifications.

1. Major Preprocessing Challenges

The primary challenge was managing co-occurring, multi-dimensional data corruption. The raw dataset contained missing values across 18 columns, extreme sentinel outliers (e.g. Annual Income recorded as 99,999,999 INR), duplicate rows, string casing noise ('Mumbay' vs 'Mumbai'), invalid age values up to 180, and scale imbalances spanning multiple orders of magnitude. Naive removal of incomplete rows would have destroyed over 80% of the dataset. Resolving these challenges required sequencing operations carefully: cleaning noise and sentinels first, performing median/mode imputation second, deduplicating third, and executing scaling/encoding last.

2. Preprocessing Step with Greatest Overall Impact

Data Cleaning (specifically Outlier Treatment via IQR Capping and Median Imputation) combined with Min-Max Normalization had the single greatest impact. Cleaning established raw statistical validity by eliminating distorting sentinels, while Min-Max Normalization scaled attributes into a uniform [0, 1] interval. Without normalization, high-variance financial metrics would completely dominate gradient descent updates and distance-based clustering. This foundational step enabled all subsequent transformations (PCA, One-Hot Encoding) to function correctly.

3. Residual Problems & Limitations

Despite rigorous preprocessing, three residual limitations remain: (1) Median/Mode imputation reduces feature variance and may smooth over subtle micro-segment signals; (2) IQR capping clips extreme values at upper bounds, which compresses legitimate ultra-high-net-worth VIP customer profiles; and (3) PCA dimensionality reduction projects 8 numerical attributes into 5 synthetic components, which sacrifices direct feature interpretability for business stakeholders.

4. Recommended Machine Learning Algorithms

For the next stage of model building, we recommend two complementary algorithms: (1) **XGBoost / LightGBM Gradient Boosted Trees** for predicting customer churn ('ChurnNextMonth') and coupon redemption ('CouponUsed'), due to their superior handling of non-linear feature interactions and categorical dummy indicators; and (2) **K-Means Clustering** applied to the 5 PCA components for customer behavioral segmentation, enabling targeted marketing campaigns.

5. Additional Preprocessing & MLOps Infrastructure before Deployment

Before deploying the pipeline into production, we recommend: (1) Implementing automated schema validation using *Great Expectations* to reject corrupted incoming payloads; (2) Building a unified, serializable Scikit-Learn Pipeline / ColumnTransformer object to prevent data leakage during inference; (3) Implementing automated bounds clipping for real-time streaming data; and (4) Establishing Population Stability Index (PSI) and Wasserstein Distance monitoring to detect feature drift over time.

Data Science Pipeline Certification: The preprocessed dataset (`cleaned_mall_customer_dataset.csv`) and fully executable Python notebook (`Assignment_1_Data_Preprocessing.ipynb`) have passed all validation checks. Missing values = 0, Duplicate rows = 0, Feature range = $[0, 1]$, Retained PCA variance = 90.4%. Pipeline ready for production model training.

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